

Definitions:

Week 1: Definition 1.1: Let A and B be two sets. We say that A is a subset of B, or that A is contained in B if and only if each $x \in A$ is also an element of B. We write $A \subseteq B$ to say that "A is a subset of B." If there is an element of A not in B, we write $A \not\subseteq B$.

Definition 1.2: The empty set, written $\emptyset$, is the set that has no elements.

Definition 1.3: If A and B are statements, their conjunction is the statement "A and B". $A \wedge B$ is true exactly when both A is true and B is true

Definition 1.4: The intersection $S \cap T$ is the set consisting of the elements S and T have in common. One can write $S \cap T = \{x : (x \in S) \wedge (x \in T)\}$

Definition 1.5: The *disjunction* of two statements A and B is the statement "A or B". $A \vee B$ is true exactly when either A is true or B is true.

Definition 1.6: The union $S \cup T$ is the set consisting of the elements of either S or T. One can write $S \cup T = \{x : (x \in S) \vee (x \in T)\}$

Definition 1.7: Let $T \subseteq S$ be a subset. The complement of T in S is the subset $S \setminus T = \{x \in S : x \notin T\}$

Definition 1.8: The converse to the conditional $A \Rightarrow B$ is the statement $B \Rightarrow A$.

Definition 1.9: The contrapositive of a conditional $A \Rightarrow B$ is the statement $\bar{B} \Rightarrow \bar{A}$

Definition 1.10: We say that two statements A and B are equivalent to mean that A is true if and only if B is true.

Definition 1.11:

Definition 2.1: A field is a set F along with operations $+$ and $\cdot$ such that the following hold for $a, b, c \in F$: Addition: A(0): $a+b \in F$, A(1): Commutativity, $a+b=b+a$, A(2): Associativity, $a+(b+c)=(a+b)+c$, A(3): There is an element $0 \in F$ such that $a+0=a$ for all $a \in F$, A(4): There is an element $-a \in F$ such that $a + (-a)=0$. Multiplication: M(0): $a \cdot b \in F$. M(1) Commutativity $a \cdot b = b \cdot a$ M(2) Associativity $a \cdot (b \cdot c) = (a \cdot b) \cdot c$ M(4) if $a \neq 0$ there exists number $1/a$ where $a \cdot 1/a = 1$. M(5) Distributivity, $a \cdot (b+c) = (a \cdot b) + (a \cdot c)$

Week 2: Definition 3.1: An ordered field is a field F, together with a relation $<$, so that the following hold: (R1) (Positive/negative) If $x \in F$, then exactly one of the following is true: $0 < x$, $x = 0$, or $x < 0$. (R2) (Negations reverse) If $y < x$, then $-x < -y$. (R3) (Can add constants) If $x < y$, and $c \in F$, then $x+c < y+c$. (R4) (Positives multiply) If $0 < x$ and $0 < y$ then $0 < xy$. (R5) (Transitive property) If $x < y$ and $y < z$ then $x < z$. **all but 2 is minimal**

Definition 3.2: For $x \in \mathbb{R}$, we define the absolute value or modulus of x to be $|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x \leq 0 \end{cases}$

Definition 4.1: Given a sequence of numbers $a_m, a_{m+1} \dots a_n$ we write $\sum_{k=m}^n a_k = a_m + a_{m+1} + \dots + a_n$

Definition 4.2: The Principle of Mathematical Induction Given a list of statements $P(k), P(k+1), \dots$, we may conclude that $P(n)$ is true for every integer $n \geq k$ provided that (1) we know that $P(k)$ is true ("base case") (2) and we can prove that $P(n) \Rightarrow P(n+1)$ for any integer $n \geq k$ ("induction step").

Week 3: Definition 5.1: Let A be a subset of $\mathbb{R}$ An upper bound for A is a real number M such that for all $x \in A$, we have $x \leq M$. We say that A is bounded above if there exists an upper bound for A, and unbounded above otherwise. • A lower bound for A is a real number m such that for all $x \in A$, we have $m \leq x$. We say that A is bounded below if there exists a lower bound for A, and unbounded below otherwise. • We say A is bounded if it is both bounded above and bounded below.

Definition 5.2: Given a subset $A \subseteq \mathbb{R}$, a number L is a least upper bound (LUB) or *supremum* for A if (a) L is an upper bound for A, that is, $x \leq L$ for all $x \in A$; and (b) $L \leq M$ for every upper bound M of A. Similarly, we say ℓ is a greatest lower bound (GLB) or *infimum* for A if (a) ℓ is a lower bound for A, that is, $x \geq \ell$ for all $x \in A$; and (b) $\ell \geq m$ for every lower bound m of A.

Definition 6.1 For $k \in \mathbb{Z}$, $(a_n)_{n=k}^{\infty}$ is defined to be a real-valued sequence beginning at index k if, for each integer $n \geq k$, the quantity a_n is a real number. For each $n \geq k$, the nth term of the sequence is defined to be a_n .

Definition 6.2 the ϵ - N definition of a limit Let $(x_n)_{n=1}^{\infty}$ be a sequence of real numbers $x_1, x_2, \dots$ and let $L \in \mathbb{R}$. We say that x_n converges to L, or L is the limit of the sequence (x_n) , if for all $\epsilon > 0$, there exists a natural number N such that for all $n > N$,

Definition 6.3 We say that a sequence (a_n) is convergent if for some $L \in \mathbb{R}$, it converges to L.